MIL-A-48028A (PA)
30 June 1975
Superseding
MIL-A-48028 (MU)
14 January 1972

MILITARY SPECIFICATION

ADAPTER-BOOSTER, BOMB, NOSE, M148E1 PARTS FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the metal parts for one type of adapter-booster designated as Adapter-Booster, Bomb, Nose, M148E1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request proposals, form a part of this specification to the extent specified herein.

*SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

FEDERAL

FED-STD-151 - Fed Test Method Std No. 151, Metal, Test Method

*MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

FSC: 1325

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MIL-STD-1169- Packaging, Packing, and Marking for Shipment of Inert Ammunition Component

DRAWINGS

U.S. ARMY PICATINNY ARSENAL

9272263 - Adapter-Booster, Bomb, Nose, Thermally Protected M148E1, Loading Assembly

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

*For meaning of asterisk see 6.5.

3. REQUIREMENTS

3.1 Material. - Material and parts shall be in accordance with applicable drawings and specifications.

3.2 Metal parts. - The metal parts shall comply with all requirements specified in Drawing (dwg.) 9272263 and associated drawings and with all requirements specified in applicable specifications.

3.3 Protective finish. - Components requiring protective finishes shall, after being subjected to 96 hours of salt spray (using a 5 percent solution of salt), show no evidence of corrosion visible to the unaided eye at normal reading distances at accidental scratches or unscratched areas of the chromate film. Corrosion products appearing at edges and corners is permissible provided that the corrosion is confined to the immediate area of the edge or corner.

3.4 Workmanship. - All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, dirt grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required marking shall be neat and sharply defined. Surface coatings shall be continuous except for a few slight scratches not exposing base material.

*3.5 First Article Inspection. - This specification contains technical provisions for first article inspection.

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Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

*4.1 Responsibility for Inspection and Standard Quality Assurance Provisions. - Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

*4.2 Classification of inspections. - The following types of inspection shall be conducted on this item.

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First Article Inspection

*4.3.1 Submission. - The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2 and 4.3.3. The first article sample shall consist of the following items in sample quantities as indicated:

Part Description	Drawings	Quantity
Adapter	9217025	10
Sleeve	9272260	10
Booster Casing Assembly	8799591	10

*4.3.2 Inspections to be performed. - See MIL-A-48078 and paras 4.4.2.1 thru 4.4.2.3 and Table I specified herein.

*4.3.3 Rejection. - (See MIL-A-48078)

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*TABLE I

FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-A-48028A (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Assembly and Components				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Adapter (Dwg. 9217025)				
	Examination for Defects	10		3.2	4.4.2.1
	Protective finish test	10		3.3	MIL-STD-151
	Sleeve (Dwg. 9272260)				
	Examination for Defects	10		3.2	4.4.2.2
	Booster Casing Assembly (Dwg. 8799591)				
	Examination for Defects	10		3.2	4.4.2.3
	Protective finish test	10		3.3	MIL-STD-151
	Air test	10		3.2	4.5.1

NOTES: The protective finish test and the air test shall be performed on the components after they have been examined for defects.

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4.4 Quality conformance inspection

*4.4.1 Inspection lot formation. - Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

*4.4.2 Examination. - (See MIL-A-48078). Unless otherwise specified in the classification of defects and test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.4.4.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-A-48028A (PA)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
*4.4.2.1	Adapter				9217025
					NEXT HIGHER ASSEMBLY
					9272262
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101	Concentricity of internal thread with large external thread		0.40%	3.2	Gage
102	Pitch diameter of large external thread		0.40%	3.2	Gage
103	Pitch diameter of small external thread		0.40%	3.2	Gage
104	Pitch diameter of internal thread		0.40%	3.2	Gage
105	Minimum length of internal full threads (when applicable)		0.40%	3.2	Gage
106	Major diameter of large external thread		0.40%	3.2	Gage
107	Major diameter of small external thread		0.40%	3.2	Gage
108	Concentricity of large external thread with small external thread at projected length		0.40%	3.2	Gage/4.4.3.2
Major B					
131	Distance small end to end of thread relief, minimum (min) (detail A2)		0.40%	3.2	Gage
132	Protective coating missing or damaged exposing base metal		0.40%	3.2	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-A-48028A (PA)

PARAGRAPH	TITLE	SHEET 2 OF 2	DRAWING NUMBER
#4.4.2.1 (cont)	Adapter		9217025
			NEXT HIGHER ASSEMBLY 9272262

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Concentricity of small internal diameter with large external thread		0.65%	3.2	Gage
202	Small inside diameter, min		0.65%	3.2	Gage
203	Total length, min		0.65%	3.2	Gage
204	Length of large outside diameter, min		0.65%	3.2	Gage
205	Minimum length of external full threads		0.65%	3.2	Gage
206	Wrench hole missing		0.65%	3.2	Visual
207	Thread relief missing (when applicable)		0.65%	3.2	Visual
208	Marking missing, misleading or unidentifiable		0.65%	3.2	Visual
209	Evidence of poor workmanship		1.00%	3.4	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-A-48028A (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.2.2	Sleeve				9272260
					NEXT HIGHER ASSEMBLY 9272262
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major B					
131	Total length, min		0.40%	3.2	Gage
132	Inside diameter, min		0.40%	3.2	Gage
133	Outside diameter, maximum (max)		0.40%	3.2	Gage
134	Small outside diameter, (max)		0.40%	3.2	Gage
135	Length of small O.D. (min)		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		1.00%	3.4	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-A-48028A (PA)

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
*4.4.2.3	Booster Casing Assembly				8799591
					NEXT HIGHER ASSEMBLY 2972262
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101	Concentricity of booster well with thread		0.40%	3.2	Gage
102	Concentricity of outside diameter with thread (see dwg. 8799592)		0.40%	3.2	Gage
103	Minor diameter of thread, max		0.40%	3.2	Gage
104	Pitch diameter of thread (see dwg 8799592)		0.40%	3.2	Gage
Major B					
131	Minimum length of full thread (see dwg. 8799592)		0.40%	3.2	Gage
132	Inside diameter of booster well, min. (see dwg. 8799593)		0.40%	3.2	Gage
133	Wall thickness (see dwg. 8799592)		0.40%	3.2	Gage
134	Inside diameter (along entire length to bottom radius), min (see dwg. 8799592)		0.40%	3.2	Gage
135	Inside diameter from 1 3/4 inches from open end to top of booster well, max (see dwg. 8799592)		0.40%	3.2	Gage
136	Protective coating missing or damaged exposing base metal		0.40%	3.2	Visual
Minor					
201	Total length, max		0.65%	3.2	Gage
202	Evidence of poor workmanship		1.00%	3.4	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-A-48028A (PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.4	Disc, Insulating		9272261
			NEXT HIGHER ASSEMBLY 9272262

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major B					
131	Diameter (max)		0.40%	3.2	Gage
132	Thickness		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		1.00%	3.4	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-A-48028A (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9272259
4.4.2.5	Washer, Insulating				NEXT HIGHER ASSEMBLY 9272262
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major B					
131	Outside diameter (max)		0.40%	3.2	Gage
132	Inside diameter (min)		0.40%	3.2	Gage
133	Thickness		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		1.00%	3.4	Visual

NOTES:

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4.4.3 Testing

*4.4.3.1 Leak test of booster casing assembly. - (See Dwg. 8799591) Major A Defect. - Three hundred and fifteen (315) samples shall be selected for this test and tested as specified in 4.5.1 using equipment in accordance with 4.4.4. If four (4) or more assemblies fail this test the lot shall be rejected. In the event one, or more, leakers are caused by anode burn-through in the fuze well the lot shall be tested 100 percent. Regardless of quality history, reduced inspection is not permitted.

*4.4.3.2 Control of adapter thread concentricity at projected length, Major B Defect: (see dwg. 9217025 and paragraph 4.4.2.1) - When the adapter is manufactured from bar stock in one set-up, i.e., the item is completely machined prior to the cut-off operation, machine control shall be determined by gaging the thread concentricity at the projected length until twenty-five (25) consecutive units are found defect free. After control has been established, and to assure that control is maintained, a sample of one (1) unit shall be randomly selected and checked from each machine from each eight (8) hours production. If any defective is found all units produced since the last acceptable check shall be rejected. The machine shall then be requalified as evidenced by twenty-five (25) consecutive defect free units being produced from that machine. Additionally, a machine shall be requalified whenever a new machine is tooled or new tooling is installed. (Non-destructive)

*4.4.3.3 Protective finish Major B defect: Adapter, and booster casing assembly. - A sample of four (4) of each component requiring protective coatings shall be selected from the inspection lot and tested as specified in Method 811.1 of FED-STD-151. The inspection lot shall be rejected if one or more components fail to conform to the specified requirements. (Destructive test)

*4.4.4 Inspection Equipment. - The inspection equipment required to perform the inspections and tests prescribed in this specification is identified in the 'Paragraph Reference/ Inspection Method' column in the tables starting with paragraph 4.4.2.1, and the test method paragraphs (see 4.5). The contractor shall submit for approval, inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078, and 6.3 herein.

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4.5 Test Methods and Procedures

4.5.1 Leak test of booster casing assembly. - The assembly shall be assembled to the applicable test equipment. It shall then be submerged in water with the brazed joint up to a minimum depth of 5/8 inch, or a soapy solution (of a consistency which will detect air leaks) may be placed over the entire brazed area. The open end shall be sealed to maintain pressure in the event of air leakage at the "O" ring. The required internal air pressure shall be applied for the specified period. Observation shall be made for the presence of a stream or streams of bubbles, of any frequency if submerged, or a foaming of the soap solution if used. The presence of either shall class a unit defective. (Non-destructive)

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking

*5.1.1 Level C. - The metal parts shall be packed separately and marked in accordance with Standard MIL-STD-1169. (Components shall not be mixed in a container.)

6. NOTES

*6.1 Intended use. - The metal parts covered by this specification are intended for use in the MK80 series Bomb.

*6.2 Ordering data. - Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Provisions for submission of first article sample.

c. Provisions of MIL-A-48078.

*6.3 Submission of Inspection Equipment Designs for Approval. (See MIL-A-48078). Submit equipment designs as required, to Commander: Picatinny Arsenal, ATTN: SARPA-QA-T-T, Dover, New Jersey 07801.

*6.4 Distribution of ammunition data cards. - Distribution of data cards shall include the following: Commander, Picatinny Arsenal, ATTN: SARPA-QA-A-M, Dover, New Jersey 07801.

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6.5 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Army-PA

Preparing Activity
Army-PA

Project Number: 1325-A-464

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NAME OF ORGANIZATION AND ADDRESS

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   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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